

Proposed Solution to #5819 SSMJ

Solution proposed by Prakash Pant, The University of Vermont, Bardiya, Nepal.

Problem proposed by Toyesh Prakash Sharma, Agra College, Agra, India

Statement of the Problem:

For $a, b, c \geq 0$, show that

$$\frac{a+b}{a^2+b^2}\sqrt{ab} + \frac{b+c}{b^2+c^2}\sqrt{bc} + \frac{c+a}{c^2+a^2}\sqrt{ca} \leq 3.$$

Solution of the Problem:

Notice that proving the following statement is enough

$$\frac{a+b}{a^2+b^2}\sqrt{ab} \leq 1 \tag{1}$$

because other two terms in L.H.S are just the same expressions with (a,b) replaced by with (b,c) or (c,a) each of which is just a generic dummy variable ≥ 0 . Thus, the total of the expression would be $\leq 1 + 1 + 1 = 3$

To prove equation (1), since $a^2 + b^2$ is positive, we can cross-multiply:

$$a^{\frac{3}{2}}b^{\frac{1}{2}} + b^{\frac{3}{2}}a^{\frac{1}{2}} \leq a^2 + b^2$$

which can be written as symmetric sums

$$\sum_{sym} a^{\frac{3}{2}}b^{\frac{1}{2}} \leq \sum_{sym} a^2b^0 \tag{2}$$

To prove this statement, we use Muirhead Inequality. Since the sequence $(\frac{3}{2}, \frac{1}{2}) \prec (2, 0)$, by muirhead inequality, equation (2) is true. This completes the proof.